Code for data preparation/visulisation to make sure that it is appropriate for stardist model

In [1]:
```
import os
os.chdir('C:/Users/User/Desktop/Results/labeling')
print(os.getcwd())
```

```
C:\Users\User\Desktop\Results\labeling
```

In [2]:
```
from __future__ import print_function, unicode_literals, absolute_import, division
import numpy as np
import matplotlib
matplotlib.rcParams["image.interpolation"] = 'none'
import matplotlib.pyplot as plt
%matplotlib inline
%config InlineBackend.figure_format = 'retina'

from glob import glob
from tqdm import tqdm
from tifffile import imread
from csbdeep.utils import Path, download_and_extract_zip_file

from stardist import fill_label_holes, relabel_image_stardist, random_label_cmap
from stardist.matching import matching_dataset

np.random.seed(42)
lbl_cmap = random_label_cmap()
```

Load in training data images

In [19]:
```
X = sorted(glob('qpproj/ground_truth/images/*.tif'))
Y = sorted(glob('qpproj/ground_truth/masks/*.tif'))
assert all(Path(x).name==Path(y).name for x,y in zip(X,Y))
```

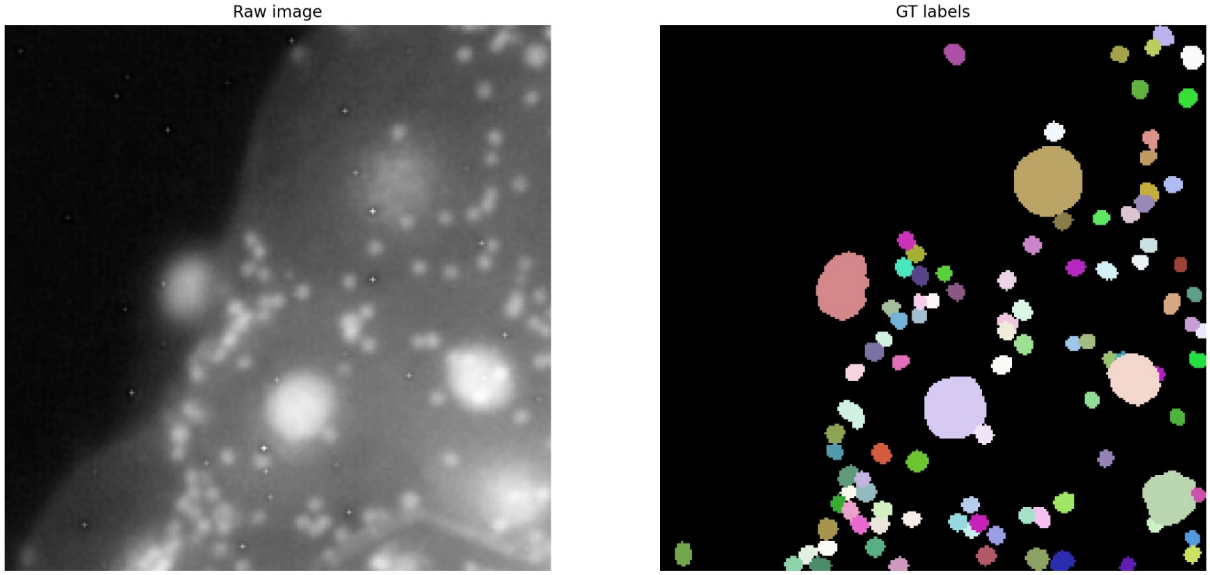
In [20]:
```
X, Y = X[:144], Y[:144]
```

In [21]:
```
X = list(map(imread,X))
Y = list(map(imread,Y))
```

In [22]:
```
i = min(4, len(X)-1)
img, lbl = X[i], fill_label_holes(Y[i])
assert img.ndim in (2,3)
img = img if img.ndim==2 else img[...,:3]
# assumed axes ordering of img and lbl is: YX(C)
```

plot example images to ensure that the labels and images are loading correctly

In [23]:
```
plt.figure(figsize=(16,10))
plt.subplot(121); plt.imshow(img,cmap='gray');   plt.axis('off'); plt.title('Raw im
plt.subplot(122); plt.imshow(lbl,cmap=lbl_cmap); plt.axis('off'); plt.title('GT lab
None;
```



Generate variable numbers of rays to describe the shapes used to label structures - this is to determine the best parameter for the model training in step 2

In [24]:
```
n_rays = [2**i for i in range(2,8)]
scores = []
for r in tqdm(n_rays):
    Y_reconstructed = [relabel_image_stardist(lbl, n_rays=r) for lbl in Y]
    mean_iou = matching_dataset(Y, Y_reconstructed, thresh=0, show_progress=False).
    scores.append(mean_iou)
```

```
100%|
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```
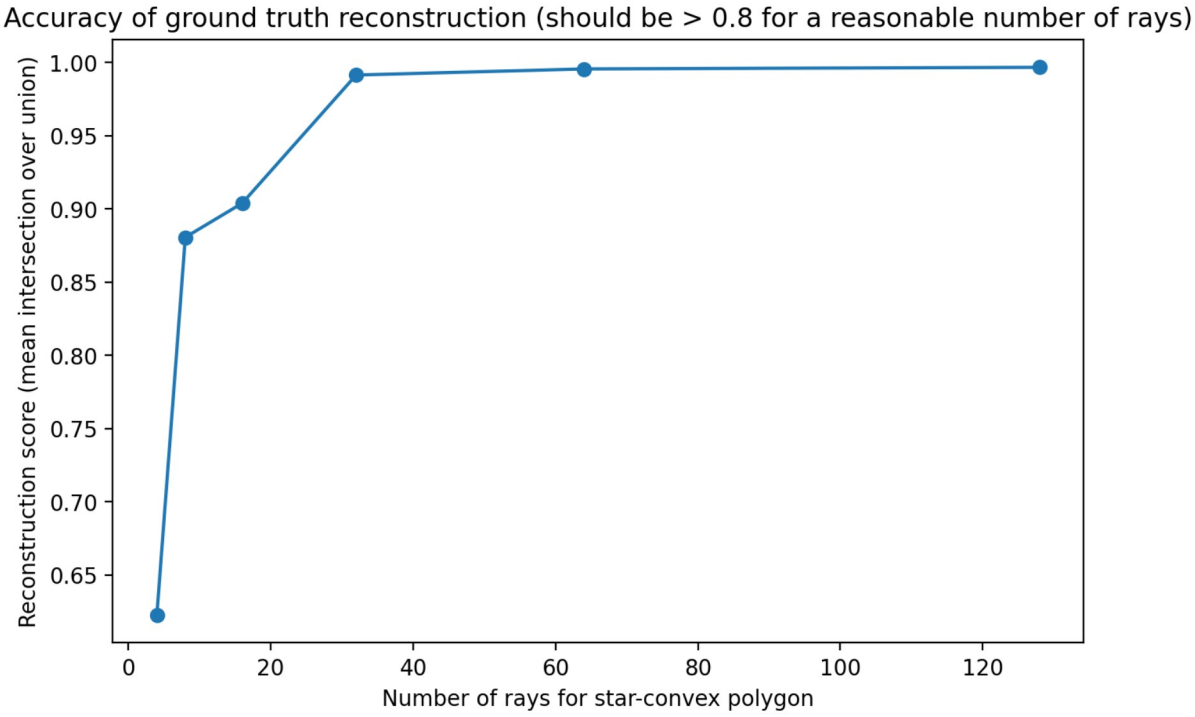
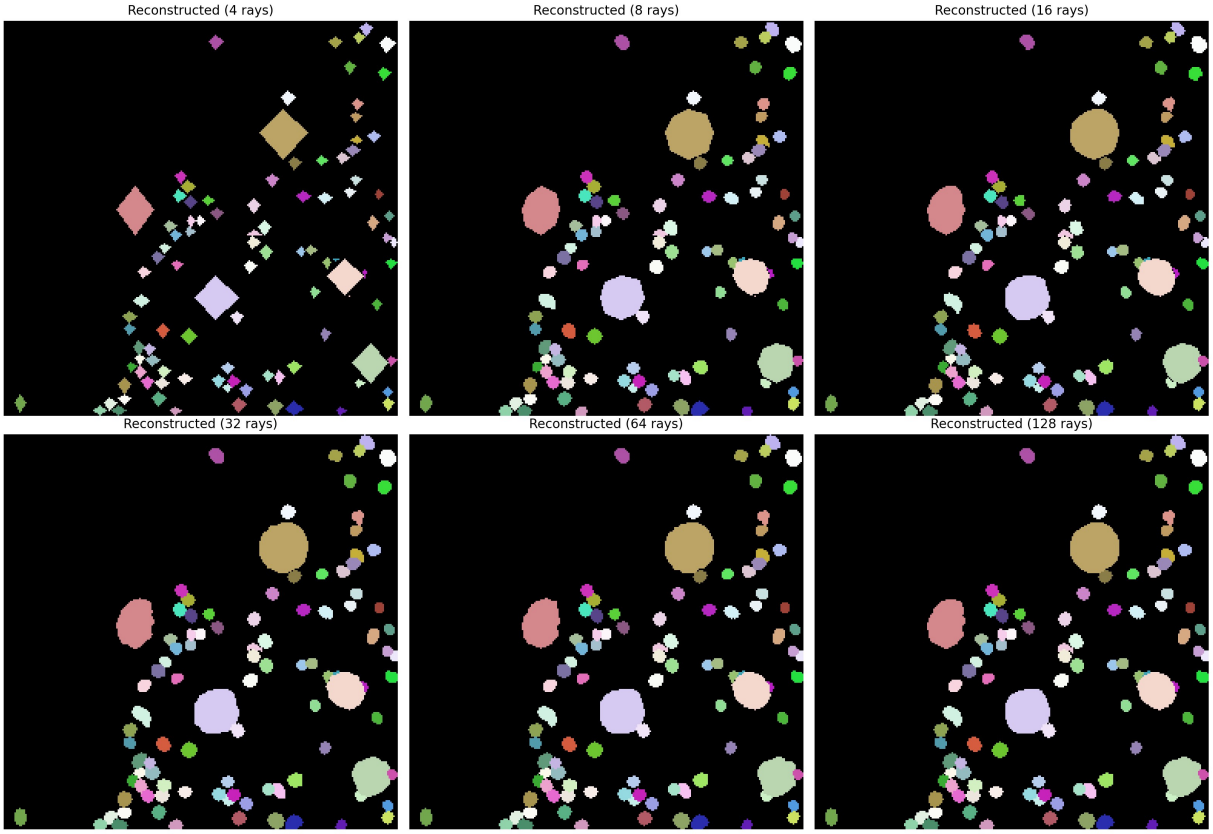

In [25]:
```
plt.figure(figsize=(8,5))
plt.plot(n_rays, scores, 'o-')
plt.xlabel('Number of rays for star-convex polygon')
plt.ylabel('Reconstruction score (mean intersection over union)')
plt.title("Accuracy of ground truth reconstruction (should be > 0.8 for a reasonabl
None;
```



example image reconstructed using variable numbers of rays showing that 32 is an appropriate number to describe the shape of the oocytes to be labeled

In [26]:
```
fig, ax = plt.subplots(2,3, figsize=(16,11))
for a,r in zip(ax.flat,n_rays):
    a.imshow(relabel_image_stardist(lbl, n_rays=r), cmap=lbl_cmap)
    a.set_title('Reconstructed (%d rays)' % r)
    a.axis('off')
plt.tight_layout();
```


Reconstructed (4 rays)
Reconstructed (8 rays)
Reconstructed (16 rays)
Reconstructed (32 rays)
Reconstructed (64 rays)
Reconstructed (128 rays)